

Ahmed Amin Ahmed Bishr

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PROFILE

AI/ML engineer with hands-on experience building production-grade systems from the ground up — from implementing algorithms from scratch to deploying full retrieval-augmented generation pipelines under real time pressure. Studying at Kafrelsheikh University, expected to graduate 2028. First-place winner of the INSTANT x ODC x Creativa AI Hackathon, where I learned RAG architecture from zero and shipped a working, evaluated clinical decision support system in under a week. Driven by a self-directed learning approach that goes beyond coursework, with experience spanning the full data science pipeline — from raw data preprocessing to Streamlit deployment and grounded LLM systems.

EDUCATION

Kafrelsheikh University

Bachelor of Computer Science — Artificial Intelligence

GPA: 3.6 / 4.00

Kafr el-Sheikh, Egypt

Expected June 2028

EXPERIENCE

IEEE Kafrelsheikh Student Branch

September 2025 – January 2026

Machine Learning Committee Intern

- Completed structured hands-on training in machine learning fundamentals through the IEEE KFS ML committee track.
- Applied supervised learning algorithms including regression and classification on real datasets as part of practical committee tasks.

Microsoft Student Club — Kafrelsheikh University

September 2025 – June 2026

Data Science Committee Intern | Technical Mentorship Program

- Completed Phase 1 of the Technical Mentorship Program (TMP) under the Microsoft Student Ambassadors Club - KFS.
- Developed foundational data science skills through a structured curriculum covering data preprocessing, exploratory data analysis, and visualization.

PROJECTS

MedLens AI — Clinical Decision Support RAG System — *Python, FastAPI, ChromaDB, Sentence-Transformers, Groq, React*

- Co-built a dual-domain clinical decision support RAG system covering dermatology conditions and Egyptian drug safety, architecting a seven-stage pipeline from data ingestion through grounded generation during a 5-day hackathon.
- Designed a per-domain embedding strategy (MiniLM for clinical guidelines, BGE-M3 for drug data) feeding independent vector stores, with cross-encoder reranking to improve retrieval relevance.
- Enforced citation-backed, grounded generation with explicit refusal logic for insufficient evidence, reducing hallucination risk in a clinical context.
- Integrated a live weather-context layer (UV index, humidity, air quality) linking environmental data to dermatology flare-up guidance.

UCMI — Used Car Market Intelligence — *Python, pandas, NumPy, Matplotlib, Seaborn, Power BI*

- Built the full data preprocessing pipeline for a used car market analytics project, handling missing values, outlier removal, type corrections, and categorical label standardization across a multi-attribute vehicle dataset.

- Ensured dataset integrity and consistency as the foundation for downstream feature engineering, EDA, and an interactive Power BI business intelligence dashboard.

Customer Churn Prediction — *Python, pandas, NumPy, scikit-learn, XGBoost, LightGBM, Streamlit*

- Built and compared multiple classification models (Logistic Regression, Random Forest, XGBoost, LightGBM, and a stacking ensemble) for a telecom customer churn dataset, achieving a best ROC-AUC of 0.936 with a tuned LightGBM model (`n_estimators=500`, `learning_rate=0.05`, `num_leaves=63`).
- Performed full ML pipeline including data preprocessing, feature engineering, class imbalance handling, hyperparameter tuning, and cross-validated model evaluation with confusion matrix, classification report, and ROC curve analysis.
- Deployed the final model as an interactive web application using Streamlit, enabling real-time churn probability prediction from user-supplied customer feature inputs.

COMPETITIONS

RAG & ROLL — 1st Place, INSTANT x ODC x Creativa AI Hackathon

August 2026

- Won 1st place among competing teams with MedLens AI, a clinical decision support RAG system, in a 5-day hackathon organized by Orange Digital Center, ITIDA, TIEC, and Creativa Innovation Hubs.
- Presented the system architecture and scalability roadmap to a judging panel of technical and industry evaluators as part of the final live demo.
- Learned Retrieval-Augmented Generation from scratch under hackathon time constraints, going from no prior exposure to a working, evaluated system in under a week.

HERE Cairo Geospatial Hackathon 2026 — HERE Technologies

February 10, 2026

- Collaborated in a 3-member team to ideate, develop, and deliver a technical solution on a real-world spatial dataset provided by HERE Technologies Cairo.
- Cleaned and preprocessed noisy GIS road data, performing feature engineering to improve data quality for downstream modeling.
- Applied machine learning techniques including regression and ensemble methods to identify patterns in geospatial data under a challenging timeframe.
- Utilized Python, pandas, NumPy, scikit-learn, Matplotlib, and Seaborn throughout the full pipeline from data ingestion to model evaluation.

CERTIFICATIONS

Certificate of Achievement — Machine Learning Committee

2026

IEEE Kafrelsheikh Student Branch

Certificate of Appreciation — Technical Mentorship Program, Phase 1 | Data Science Track

2026

Microsoft Student Ambassadors Club — KFS

SKILLS

Machine Learning & Generative AI: Retrieval-Augmented Generation (RAG), LLM Inference (Groq), Embedding Models (Sentence-Transformers, MiniLM, BGE-M3), Vector Databases (ChromaDB), Cross-Encoder Reranking, Grounded Generation & Citation Design, Linear/Logistic Regression, Decision Trees, Random Forest, XGBoost, LightGBM, Model Evaluation & Deployment

Software & Data Engineering: Python, C++, SQL, FastAPI, pandas, NumPy, scikit-learn, OpenCV, Git, GitHub

Data Analysis & Visualization: Exploratory Data Analysis, Feature Engineering, Data Preprocessing, Cross-Validation, Matplotlib, Seaborn

Soft Skills: Teamwork, Communication, Performance Under Pressure, Problem Solving, Self-Directed Learning